

●INTERVIEW PREP 2025

POWER QUERY

INTERVIEW Q&A

55 questions across 5 sections — from basics to scenario-based problems. For DA & BA roles at TCS, Deloitte, Amazon & more.

55+

QUESTIONS

30

TOPICS

5

SECTIONS

VR

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ANALYTICS

SAMPLE QUESTIONS INSIDE ↓

CONCEPT

What is the difference between Merge and Append Queries?

★ MUST KNOW

JOIN TYPES

Explain all 6 join types in Power Query with real examples.

★ MUST KNOW

TRANSFORMATION

What is Pivot vs Unpivot? Why does Power BI prefer long format data?

IMPORTANT

QUERY MANAGEMENT

Difference between Reference Query and Duplicate Query?

IMPORTANT

SCENARIO BASED

Dataset has 6 metadata rows at top & 3 footer rows. How do you clean it?

SCENARIO

SECTION 1 — Basics & Interface (Q1–Q8)

Q1. What is Power Query and why do we use it in Power BI?

A: Power Query is Power BI's data transformation engine (also called Get & Transform). It allows us to connect to data sources, clean, reshape, and combine data before it loads into the data model. Every transformation step is recorded and auto-replayed on refresh — so the model always has clean, updated data without manual intervention.

Interview Tip: Always mention 'auto-replay on refresh' — this shows you understand why Power Query is better than manual Excel cleaning.

Q2. What are the 4 main areas of the Power Query Editor?

A: 1. Queries Pane (left) — lists all loaded tables. 2. Data Preview (center) — shows rows and columns of the selected table. 3. Ribbon (top) — all transformation buttons (Home, Transform, Add Column, View tabs). 4. Applied Steps (right) — records every transformation step in order.

Interview Tip: Draw this on paper during interview if asked — it shows confidence.

Q3. What is the Applied Steps pane and why is it important?

A: Applied Steps records every transformation you apply as a named step — like 'Removed Columns', 'Changed Type', 'Filtered Rows'. You can click any step to see the data at that point, reorder steps, delete steps, or edit the M code behind them. This makes Power Query fully auditable and repeatable.

Interview Tip: Mention that each step generates M code automatically — this impresses interviewers.

Q4. What is M language in Power Query?

A: M (also called Power Query Formula Language) is the functional programming language that runs behind Power Query. Every time you click a button in the UI — Trim, Filter, Group By — Power Query writes M code automatically. Advanced users can write M directly in the Advanced Editor for complex transformations that the UI cannot handle.

```
// Example M code for a simple filter = Table.SelectRows("#Changed Type", each [Region] = "North")
```

Q5. What is the difference between Transform tab and Add Column tab?

A: Transform tab modifies the existing column in place — the original column is replaced. Add Column tab creates a new column while keeping the original column unchanged. For example, Text.Upper in Transform tab overwrites the column. Text.Upper in Add Column tab adds a new column next to the original.

Interview Tip: This is a very common interview question — always give a concrete example.

Q6. What data types are available in Power Query?

A: Text, Whole Number, Decimal Number, Fixed Decimal Number, Percentage, Date, Time, Date/Time, Date/Time/Timezone, Duration, True/False (Boolean), Binary, and Any. 'Any' type means Power Query hasn't determined the type — always change Any to the correct specific type.

Note: 'Any' data type causes calculation errors in DAX — always fix it before closing Power Query.

Q7. What is Query Folding in Power Query?

A: Query Folding is when Power Query pushes transformation steps back to the data source (like SQL Server) to execute there instead of in Power BI. This improves performance because the source processes the data before sending it. Folding works best with database sources. Excel and CSV sources generally don't support query folding.

Interview Tip: Mention query folding when asked about Power Query performance optimization.

Q8. How is Power Query different from Excel formulas?

A: Power Query: handles millions of rows, auto-refreshes, doesn't modify source data, records steps as replayable M code, combines multiple sources easily. Excel formulas: work cell by cell, break on large data, require manual updates, modify the actual spreadsheet. Power Query is a data preparation layer — Excel formulas are calculations on already-loaded data.

SECTION 2 — Data Cleaning (Q9–Q20)

Q9. What is the difference between Trim and Clean in Power Query?

A: Trim removes leading spaces, trailing spaces, and extra spaces between words. Clean removes non-printable invisible characters like line breaks, tab characters, and control characters. Best practice is to apply both together on text columns — Trim first, then Clean.

Interview Tip: In interviews say: 'I always apply both Trim and Clean together on text columns to handle all types of hidden characters.'

Q10. What is the difference between Replace Values and Replace Errors?

A: Replace Values replaces a specific known text or number with another value — like replacing 'electronics' with 'Electronics'. Replace Errors specifically targets red error cells (cells where Power Query encountered a processing error) and substitutes them with a safe value like null or 0.

Interview Tip: Replace Errors is used after changing data types — when some cells can't convert, they become errors.

Q11. What is the difference between Remove Errors and Keep Errors?

A: Remove Errors deletes entire rows that contain an error in the selected column — use when error rows are invalid data. Keep Errors shows only the error rows — use when you want to investigate what went wrong. Keep Errors is a data quality check tool, not a cleaning tool.

Q12. What is the difference between Remove Duplicates on one column vs multiple columns?

A: Single column: removes rows where that one column value repeats — keeps first occurrence. Multiple columns: removes rows only when ALL selected columns are identical together — more precise. For example, same InvoiceNo with different CustomerName is NOT a duplicate in multi-column mode.

Interview Tip: Always clarify with the interviewer — 'duplicate based on which columns?' — shows analytical thinking.

Q13. What is the difference between Fill Down and Fill Up?

A: Fill Down copies the last non-blank value downward to fill blank cells below it. Fill Up copies the next non-blank value upward to fill blank cells above it. Simple rule: blank cells are below the value → Fill Down. Blank cells are above the value → Fill Up. Fill Down is used in 90% of real cases (ERP/SAP exports).

Interview Tip: Give an example: 'In SAP exports, department name appears once at the top and blanks below — Fill Down fills those blanks.'

Q14. Why should you use null instead of 0 when replacing errors?

A: 0 implies the value is zero — for example, zero sales or zero quantity. null means the value is unknown or missing. Replacing errors with 0 creates false data that distorts averages, totals, and ratios. null is handled correctly by DAX — it's excluded from averages and shown as blank in visuals.

Note: Never replace amount errors with 0 in financial data — always use null.

Q15. What does 'Detect Data Type' do?

A: Detect Data Type automatically assigns the most appropriate data type to each column based on the values it finds. It's useful after Promote Headers or after Append operations where all columns reset to Text type. However, it can sometimes guess wrong — always verify after using it.

Q16. How do you handle a column with mixed date formats like MM/DD/YYYY and DD-MM-YYYY?

A: First change the data type using locale — right click the column header → Change Type → Using Locale → select Date and choose the correct locale (English US for MM/DD/YYYY, English India for DD-MM-YYYY). If formats are truly mixed in the same column, use a Custom Column with try-otherwise M code to handle each format.

Interview Tip: Mentioning 'Change Type Using Locale' shows real-world experience.

Q17. What is Column Quality, Column Distribution, and Column Profile?

A: Column Quality (View tab): shows % Valid, Error, and Empty per column as colored bars. Column Distribution: shows a histogram of value frequency for each column. Column Profile: shows deep statistics for the selected column — Count, Min, Max, Average, Standard Deviation, Distinct count, Unique count. All three are under the View tab.

Interview Tip: By default Column Profile shows only first 1000 rows — change to 'entire dataset' for accurate stats.

Q18. What is the difference between Distinct count and Unique count in Column Profile?

A: Distinct count = number of different values in the column (e.g., 5 regions = 5 distinct). Unique count = number of values that appear exactly once. If 'North' appears 50 times, it counts as 1 distinct but 0 unique. If 'Special Region' appears only once, it counts as 1 distinct AND 1 unique.

Q19. How do you remove metadata rows from a system-exported file?

A: Step 1: Home tab → Remove Rows → Remove Top Rows → enter number of metadata rows. Step 2: Home tab → Use First Row as Headers (promotes actual header row). Step 3: Home tab → Remove Rows → Remove Bottom Rows → remove footer rows. Step 4: Detect Data Type to fix column types.

Interview Tip: This exact 4-step flow is used for every SAP, Oracle, and ERP export — mention it confidently.

Q20. What happens when you change data type on a column that has incompatible values?

A: Power Query converts values it can handle and marks incompatible values as errors (shown as red cells with the word 'Error'). For example, changing a column with mostly numbers but some text like 'ERROR' to Decimal Number — the text cells become error cells. You then handle these with Replace Errors or Remove Errors.

SECTION 3 — Transformations (Q21–Q35)

Q21. What is the difference between Split Column by Delimiter and by Number of Characters?

A: By Delimiter splits where a specific character appears — like splitting 'ELEC-LAP-001' by '-'. By Number of Characters splits after a fixed number of characters regardless of content — like splitting a PAN number 'ABCDE1234F' into 5+4+1 characters. Use Delimiter for separated data, Number of Characters for fixed-width data.

Q22. What is the difference between Merge Columns and Add Column → Merge Columns?

A: Transform tab → Merge Columns replaces the selected columns with one combined column — originals are deleted. Add Column tab → Merge Columns creates a new combined column while keeping the originals intact. Always use Add Column tab if you need to keep the original columns.

Interview Tip: This applies to all operations — Transform tab modifies, Add Column tab adds new.

Q23. What Extract operations are available in Power Query?

A: First Characters (first N chars), Last Characters (last N chars), Range (start position + length), Text Before Delimiter, Text After Delimiter, Text Between Delimiters, Length (character count of the value). All available under Add Column tab → Extract.

Interview Tip: Give a real example: 'Extract Text After @ to get email domain from email column.'

Q24. What is the difference between Pivot and Unpivot?

A: Pivot converts unique row values into column headers — turns long format into wide format. Unpivot converts column headers into row values — turns wide format into long format. Pivot: more columns, fewer rows. Unpivot: fewer columns, more rows. Power BI prefers long format for analysis.

Interview Tip: This is one of the top 5 most asked Power Query questions. Always have a clear answer ready.

Q25. Why does Power BI prefer long (unpivoted) format over wide format?

A: Power BI visuals, slicers, and DAX work on rows — not column headers. To filter by Month in a slicer, Month must be a value in a row. If months are column headers (wide format), you cannot slice by them. Unpivoting converts column headers into rows — making the data compatible with all Power BI features.

Interview Tip: Say this in interviews: 'Wide format is for human reading. Long format is for machine analysis.'

Q26. What is the difference between Unpivot Columns and Unpivot Other Columns?

A: Unpivot Columns: select the columns you WANT to unpivot → those get converted to rows. Unpivot Other Columns: select the columns you WANT TO KEEP → everything else gets unpivoted. When you have many columns to unpivot, Unpivot Other Columns is faster — select the 2-3 identity columns to keep and let the rest unpivot automatically.

Q27. What is Transpose in Power Query and when do you use it?

A: Transpose flips the entire table — rows become columns and columns become rows. Used when data arrives rotated the wrong way from a system export. After transposing, headers are lost (become Column1, Column2) so you must always follow with Use First Row as Headers. Transpose is less common than Unpivot in real projects.

Q28. How does Group By work in Power Query?

A: Group By summarizes data by grouping rows with the same value in a key column and computing aggregations (Sum, Count, Average, Min, Max) on other columns. Basic mode: one aggregation. Advanced mode: multiple aggregations in one step. Result has one row per unique group value.

Interview Tip: Advanced Group By with multiple aggregations in one step is a strong answer that shows experience.

Q29. What aggregation functions are available in Group By?

A: Sum, Average, Median, Min, Max, Count Rows, Count Distinct Rows, All Rows (returns a nested table). Count Rows counts total rows in the group. Count Distinct Rows counts unique values. All Rows is used for advanced nested table operations in M.

Q30. What is a Conditional Column and how is it different from a Custom Column?

A: Conditional Column: point-and-click UI for IF/ELSE logic — no code needed — limited to simple conditions. Custom Column: write M code directly — handles complex calculations, text operations, nested conditions, function calls. For simple IF/THEN use Conditional Column. For calculations like profit margin or combining text use Custom Column.

Q31. What are the 3 types of Index Column?

A: From 0 (starts at 0, increments by 1 — programming style). From 1 (starts at 1, increments by 1 — most common in reports). Custom (you set the starting number and increment value — e.g., start at 1000, increment by 5 gives 1000, 1005, 1010...).

Q32. What number transformations are available in Power Query?

A: Rounding: Round, Round Up, Round Down. Standard arithmetic: Add, Subtract, Multiply, Divide, Modulo, Percentage. Scientific: Absolute Value, Square Root, Power, Factorial, Logarithm. Information: Is Even, Is Odd. All available under Transform tab when a number column is selected.

Q33. What Date and Time transformations are available in Power Query?

A: Date: Year, Month Number, Month Name, Day, Day Name, Day of Week, Quarter, Week of Year, Start/End of Month, Start/End of Quarter, Start/End of Year, Age. Time: Hour, Minute, Second, Start/End of Hour. Duration: Total Days, Total Hours, Total Minutes, Total Seconds. Date.Age gives duration since a date — divide by 365 for years.

Interview Tip: 'Date.Age divided by 365 to get age in years' shows practical experience.

Q34. What is Add Prefix and Add Suffix used for?

A: Add Prefix adds standard text to the beginning of every value in a column — e.g., adding 'EMP-' before every employee ID. Add Suffix adds text to the end — e.g., adding '-IND' after every country code. Both are under Transform tab → Format. Use Add Column tab version to keep originals.

Q35. How do you combine First Name and Last Name columns in Power Query?

A: Method 1: Select both columns → Transform tab → Merge Columns → choose Space as separator → name the new column 'FullName' (this deletes originals). Method 2: Add Column tab → Custom Column → formula: [FirstName] & " " & [LastName] (this keeps originals). Method 3: Add Column tab → Merge Columns (keeps originals).

Interview Tip: Always mention both methods and when to use each — shows depth of knowledge.

SECTION 4 — Merge, Append & Query Management (Q36–Q47)

Q36. Explain all 6 join types in Power Query Merge.

A: Power Query supports 6 join types — each returns a different subset of rows:

Join Type	Returns
Left Outer	All rows from Left table + matching from Right. Non-matching Right = null
Right Outer	Matching from Left + all rows from Right. Non-matching Left = null
Full Outer	ALL rows from BOTH tables. Non-matching on either side = null
Inner	ONLY rows that match in BOTH tables. Non-matching excluded
Left Anti	Rows in Left that have NO match in Right (orphan records)
Right Anti	Rows in Right that have NO match in Left (unmatched records)

Interview Tip: Use Left Anti to find orphan records (data quality check). Use Inner for clean joined reports.

Q37. What is the difference between Merge Queries and Append Queries?

A: Merge Queries combines tables horizontally — adds columns from one table to another based on a matching key (like SQL JOIN). Result has same or fewer rows, more columns. Append Queries combines tables vertically — stacks rows from multiple tables on top of each other (like SQL UNION). Result has more rows, same columns.

Interview Tip: 'Merge = JOIN, Append = UNION' — the SQL equivalent analogy impresses interviewers.

Q38. What happens when you Append tables with different column names?

A: Power Query creates a union of all columns from both tables. Rows from Table A get null in columns that only exist in Table B, and vice versa. This is why column names must match exactly before appending — 'Total Amount' and 'TotalAmount' (with/without space) are treated as two different columns.

Note: Column name mismatch is the #1 reason Append produces unexpected null values.

Q39. What is the difference between Append Queries and Append Queries as New?

A: Append Queries: modifies the current table directly — adds rows to it permanently. Append Queries as New: creates a brand new query combining both tables — originals remain unchanged. Always use 'as New' in practice — keeps source tables clean and gives you a separate combined query to work with.

Q40. What is the difference between Reference Query and Duplicate Query?

A: Duplicate: creates a fully independent copy of the query with all its steps — changes to original don't affect the duplicate. Reference: creates a linked query that points to and reads from the original — if original changes, reference automatically reflects those changes. Reference = SQL View. Duplicate = SQL table copy.

Interview Tip: Reference query has fewer Applied Steps (just one pointing to source) — more efficient and maintainable.

Q41. What is Fuzzy Merge and when do you use it?

A: Fuzzy Merge joins tables on approximate matches instead of exact matches — handles typos, extra spaces, different casing, and name variants. Used when joining on name or address columns from different data sources that may have data entry inconsistencies. Threshold of 0.8 (80% similarity) is the recommended starting point.

Interview Tip: 'Fuzzy Merge is used for data quality issues in name/address matching — threshold 0.8 is the sweet spot' is a strong interview answer.

Q42. How does Fuzzy Merge threshold work?

A: Threshold range is 0.0 to 1.0. 1.0 = exact match only (same as regular merge). 0.8 = 80% similarity — catches most typos and formatting differences. 0.6 = 60% similarity — too lenient, may produce wrong matches. Lower threshold = more matches but more false positives. Higher threshold = fewer matches but more accurate.

Q43. How do you merge on multiple columns (compound key)?

A: In the Merge dialog — hold Ctrl and click multiple columns in both the left and right tables. Power Query highlights them in numbered order (1, 2, 3...) showing the join order. The merge matches rows where ALL selected columns match together — more precise than single column merge.

Interview Tip: Compound key merges are common in real projects — Order + Product + Date together as the unique key.

Q44. What are Parameters in Power Query and why are they useful?

A: Parameters are named variables stored in Power Query that can be used inside filter steps, file paths, or any M expression. Instead of hardcoding values like 'North' in a filter — store it as a parameter. When the value changes, update only the parameter — all queries using it update automatically. Useful for date range filters, region selection, and dynamic file paths.

Q45. What is the difference between Merge Queries and Merge Queries as New?

A: Merge Queries: modifies the current query — adds merged columns directly to it. Merge Queries as New: creates a new query with the merged result — both originals remain unchanged. Use 'as New' when you want to keep the source tables clean for other uses.

Q46. How do you expand columns after a Merge?

A: After merging, the right table appears as a single column with a table icon. Click the expand icon (double arrow ) on that column header → a dialog appears showing all columns from the right table → select which columns to bring in → uncheck 'Use original column name as prefix' for cleaner names → click OK.

Interview Tip: Unchecking 'Use original column name as prefix' is important — otherwise column names become long like 'Customers.City'.

Q47. How do you combine multiple Excel files from a folder in Power Query?

A: Home tab → New Source → More → Folder → browse to the folder → click OK → Power Query lists all files → click Combine → Combine and Transform → select the sheet to combine from. Power Query creates a custom function that applies the same transformation to all files and appends them into one table.

Interview Tip: This is heavily used in real projects where monthly data files are stored in one folder.

SECTION 5 — Scenario Based & Advanced (Q48–Q55)

Q48. SCENARIO: Your dataset has 6 metadata rows at top and 3 footer rows at bottom. How do you clean it?

A: Step 1: Home tab → Remove Rows → Remove Top Rows → enter 6. Step 2: Home tab → Use First Row as Headers. Step 3: Home tab → Remove Rows → Remove Bottom Rows → enter 3. Step 4: Home tab → Detect Data Type (or manually set types). This is the standard ERP/SAP export cleanup workflow.

Interview Tip: Knowing this exact 4-step workflow is expected in any DA/BA interview using Power BI.

Q49. SCENARIO: You have sales data in 12 separate monthly Excel files. How do you combine them?

A: Use Combine Files from Folder feature: New Source → Folder → select folder → Combine → Combine and Transform → select sheet. Power Query auto-generates a helper function, applies it to all 12 files, and appends results into one table. When a 13th month file is added to the folder, simply refresh — it's automatically included.

Q50. SCENARIO: You have a wide table with Jan, Feb, Mar...Dec as columns. Power BI slicer can't filter by month. How do you fix it?

A: Unpivot the month columns: select Jan through Dec → Transform tab → Unpivot Columns → rename Attribute to 'Month' and Value to 'Amount'. Now Month is a column with values Jan, Feb etc — each row is one month per record. Power BI slicer can now filter by Month correctly.

Interview Tip: Wide → Long format conversion using Unpivot is the answer to many 'my slicer doesn't work' problems.

Q51. SCENARIO: Two tables have customer names but with typos — 'Ravi Kumar' vs 'Ravi Kumar'. Regular merge gives no match. What do you do?

A: Use Fuzzy Merge: Home tab → Merge Queries → enable 'Use fuzzy matching to perform the merge' → set threshold to 0.8 → merge on the CustomerName column. Power Query uses text similarity algorithms to match 'Ravi Kumar' with 'Ravi Kumar' (extra space) as the same customer.

Q52. SCENARIO: You need to create 3 filtered views of the same Sales table — Completed, Cancelled, Pending — without loading data 3 times. How?

A: Use Reference Query: right click Sales table → Reference → creates Sales (2) → rename to Sales_Completed → filter Status = Completed. Repeat for Cancelled and Pending. All three reference queries read from the original Sales table — data is loaded once, all 3 views update automatically when Sales table changes.

Q53. SCENARIO: After merging Orders with Customers using CustomerID, some orders show null in customer columns. What does this mean and how do you investigate?

A: Nulls in customer columns after a Left Outer Join mean those orders have CustomerIDs that don't exist in the Customers table — orphan records. To investigate: redo the merge with Left Anti join — this shows only the unmatched order rows. Check their CustomerIDs — they may be invalid, newly added, or data entry errors.

Interview Tip: Left Anti join for data quality investigation is a strong answer that shows real-world thinking.

Q54. What is a Custom Function in Power Query and how do you create one?

A: A Custom Function is reusable M code that accepts inputs and returns an output — like a formula you write once and apply anywhere. Create it: Home tab → New Source → Blank Query → Advanced Editor → write the function → Done. Use it in any table via Add Column → Custom Column → call the function name with parameters.

```
// Example: Tax calculator function (amount as number, taxRate as number) => amount * taxRate // Use in Custom Column: fnTaxCalculator([Amount], 0.18)
```

Q55. What is the let...in expression in M language?

A: let...in is the structure of every M query. 'let' defines named steps (variables) one after another. 'in' specifies which step's result to return as the final output. Each step can reference previous steps by name. This is what the Applied Steps pane generates automatically behind the scenes.

```
let Source = Excel.CurrentWorkbook(){[Name="Sales"]}[Content], Filtered = Table.SelectRows(Source, each [Status] = "Completed"), Sorted = Table.Sort(Filter, {"Amount", 1}) in Sorted
```

QUICK REFERENCE — Most Asked Topics

#	Topic	One Line Answer
1	Merge vs Append	Merge = JOIN (horizontal). Append = UNION (vertical)
2	All 6 Join Types	Left Outer, Right Outer, Full Outer, Inner, Left Anti, Right Anti
3	Pivot vs Unpivot	Pivot = long to wide. Unpivot = wide to long. PBI prefers long.
4	Reference vs Duplicate	Reference = linked (auto-updates). Duplicate = independent copy.
5	Fill Down vs Fill Up	Blanks below value = Fill Down. Blanks above value = Fill Up.
6	Transform vs Add Column	Transform = modifies existing. Add Column = keeps original.
7	Fuzzy Merge	For name/address typos. Threshold 0.8 recommended.
8	Promote Headers	Remove Top Rows → Use First Row as Headers → Remove Bottom Rows
9	Why Unpivot for PBI?	PBI visuals work on rows not column headers. Long format needed.
10	Parameters	Named variables for dynamic filters. Change once → all queries update.

Final Interview Advice: Power Query questions are 30% conceptual (what is X) and 70% scenario-based (how would you handle Y). Practice explaining your answers with the dataset you built — mention specific sheets like Sales_Raw, MJ_Orders, Targets_Monthly. Real examples beat textbook definitions every time. Good luck, Venkatesh!